

**The Federal State Autonomous Educational Institution of Higher Education  
"Peoples' Friendship University of Russia named after Patrice Lumumba"**

**ANNOTATION  
of the Graduation Qualification Paper**

**by Zhendorenko Ivan Borisovich**

**subject: « *Designing intelligent autonomous agents for diagnostics and self-healing of distributed systems* »**

*This final qualifying thesis is devoted to the design of intelligent autonomous agents for the diagnostics and self-healing of distributed systems.*

*The thesis consists of an introduction, theoretical and practical sections, a conclusion, and a list of references.*

*The introduction substantiates the relevance of using agent-based systems for incident detection and prevention. It defines the goal, objectives, object, and subject of the study, as well as the theoretical and methodological foundations of the work.*

*The object of the study is distributed computing systems and the processes for their monitoring, diagnostics, and recovery. The subject of the study is methods and algorithms for constructing and orchestrating intelligent autonomous agents that use observability data to make management decisions under uncertainty.*

*The goal of the thesis is to develop the architecture and algorithms for an intelligent agent system that provides automatic anomaly detection, incident cause diagnosis, and the secure execution of self-healing actions for distributed services based on observability data. To achieve this goal, the following objectives were set:*

*1) Analyze the theoretical foundations of distributed computing systems, their architectures, and the principles of building resilient systems;*

*2) Explore the concept of observability as the foundation for diagnosing distributed systems, and analyze modern approaches to collecting and analyzing metrics, logs, and traces;*

3) Conduct a comparative analysis of backend approaches to building resilient systems: synchronous and asynchronous architectures, REST and gRPC, event-driven systems, CQRS, principles of idempotency, and resilience patterns.

4) Study the engineering paradigms of DevOps and SRE, deployment automation methods, self-healing strategies in Kubernetes, and the limitations of modern monitoring systems.

5) Explore anomaly detection and degradation prediction methods: statistical methods, machine learning methods for time series, and the use of LLM for log analysis.

6) Analyze the architectures of intelligent autonomous agents. 7) Implement a prototype of an intelligent autonomous agent system, develop the module structure, interaction schemes, configuration parameters, and simulate self-healing scenarios;

8) Conduct an experimental study of the effectiveness of the developed approach: develop an experimental methodology, define test scenarios, and evaluate metrics (MTTR, latency, precision, recall), and analyze the results.

The theoretical section examines methods and approaches to distributed system architecture, anomaly detection, and incident prevention, and also examines the limitations of existing stacks.

The practical section describes the development of a multi-agent system, specifically the development of an event-driven architecture, the main system modules, an agent orchestration mechanism, system integration with LLM, scenario development, and system testing.

The conclusion contains general findings from the study. The main finding is that the developed multi-agent system effectively handles recovery scenarios.

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